

Homework 4

- 1 The solution to *IIB* string theory with Q1 D1 brane and Q5 D5 brane can be described by a black string solution in 6d, with the metric

$$\begin{aligned}
 ds^2 = & (f_1 f_5)^{-1/2} (-dt^2 + d\phi^2 + \frac{r_0^2}{r^2} (\cosh \sigma dt + \sinh \sigma d\phi)^2) \\
 & + (f_1 f_5)^{1/2} (\frac{dr^2}{1 - r_0^2/r^2} + r^2 \Omega_3^2)
 \end{aligned} \tag{1}$$

where

$$f_1 = 1 + \frac{r_1^2}{r^2}, \quad f_5 = 1 + \frac{r_5^2}{r^2} \tag{2}$$

and $\phi \sim \phi + 2\pi R$.

- a Find the event horizon.
- b Find the near horizon geometry of the extremal black hole. *hint:*

$$r_0 = 0, \quad r \rightarrow \lambda r, \quad t \rightarrow t l / \lambda, \quad \phi \rightarrow \phi l / \lambda. \tag{3}$$

- c Find the near horizon geometry of the near extremal black hole. *hint:*

$$r \rightarrow \lambda r, \quad t \rightarrow t l / \lambda, \quad \phi \rightarrow \phi l / \lambda, \quad r_0 \rightarrow \lambda r_0. \tag{4}$$

- d Calculate the Hawking temperature and the chemical potential. *hint: expand in the near horizon region $r = r_h + \epsilon$, it must have the structure $ds^2 = -a\epsilon^2 dt^2 + b d\epsilon^2 + c(d\phi - \Omega_H dt)^2 + \dots = -a\epsilon^2 dt^2 + b d\epsilon^2 + c(d\phi')^2 + \dots$. Comparing with Rindler spacetime or considering the Euclidean geometry, the identification should be $(t, \phi) \sim (t + i\beta, \phi + i\beta\Omega_H)$.*